

Motor & Driver Comparison Guide

Line Follower Robot — Complete Reference

1 · Motor Overview

N20 Gear Motor		★ BEST FOR MOST BUILDS	
Type	Brushed DC (Geared)	Voltage	3 V – 12 V
RPM Range	100 – 1000 RPM (geared)	Torque	Medium (gear reduction)
Size	Very small	Control	Simple PWM + H-bridge
Best For	General / Beginner line followers		
Pros	Cheap · Easy to control · Many RPM options · Compact		
Cons	Brushes wear · Less efficient at very high speeds		

Coreless Motor		★ BEST FOR SPEED COMPETITION	
Type	Brushed DC (High-Speed)	Voltage	3 V – 7.4 V
RPM Range	10,000 – 50,000 RPM (before gearing)	Torque	Low (needs gearbox)
Size	Small	Control	Fast-PWM H-bridge
Best For	High-speed competition bots		
Pros	Low inertia · Fast response · Smooth		
Cons	Needs gearbox · Delicate · Pricier		

16 GA DC Motor		SUITABLE FOR MID-SIZE BOTS	
Type	Brushed DC (Mid-size)	Voltage	6 V – 12 V
RPM Range	100 – 600 RPM	Torque	Medium-High
Size	Medium	Control	Standard H-bridge
Best For	Mid-weight robots (200–400 g)		
Pros	Good torque · Reliable · Widely available		
Cons	Bulkier than N20 · Higher current draw		

12 GA DC Motor		OVERKILL FOR LINE FOLLOWERS	
Type	Brushed DC (Heavy-Duty)	Voltage	6 V – 24 V
RPM Range	50 – 300 RPM	Torque	High
Size	Large	Control	High-current H-bridge
Best For	Heavy / Industrial robots		
Pros	Very high torque · Durable		
Cons	Too large for typical line followers		

Micro BLDC Motor		ADVANCED / COMPETITION BUILDS	
Type	Brushless DC (3-phase)	Voltage	3.3 V – 12 V
RPM Range	5,000 – 100,000 RPM	Torque	Low–Medium
Size	Very small – small	Control	3-phase driver / ESC
Best For	Advanced high-speed competition		
Pros	Highly efficient · Long life · No brush wear		
Cons	Complex driver · Higher cost · Needs tuning		

2 · Motor Driver Details

N20 Gear Motor · Brushed DC			
Driver IC	Voltage	Current	Notes
L298N	4.5–46 V	2 A / ch	Dual H-bridge, most common, easy wiring
L293D	4.5–36 V	0.6 A / ch	Lightweight, good for 3.3 V/5 V systems
TB6612FNG	2.5–13.5 V	1.2 A / ch	Low heat, efficient, better than L298N
DRV8833	2.7–10.8 V	1.5 A / ch	Very compact, ideal for tiny bots
MX1508	2–10 V	1.5 A / ch	Ultra-small, made for N20 specifically

Recommended: TB6612FNG or DRV8833

Coreless Motor · High-Speed Brushed			
Driver IC	Voltage	Current	Notes
TB6612FNG	2.5–13.5 V	1.2 A / ch	Fast PWM response — top pick
DRV8833	2.7–10.8 V	1.5 A / ch	Handles fast switching, low dropout
DRV8871	6.5–45 V	3.6 A	Good current handling for coreless
L293D	4.5–36 V	0.6 A / ch	Works but not ideal at very high speeds
MX1508	2–10 V	1.5 A / ch	Works for small coreless motors

Recommended: TB6612FNG or DRV8833

16 GA DC Motor · Mid-Size Brushed

Driver IC	Voltage	Current	Notes
L298N	4.5–46 V	2 A / ch	Handles higher current well
BTS7960 (IBT-2)	5.5–27 V	43 A peak	Excellent for higher current motors
DRV8701	5.9–45 V	15 A	High current, industrial grade
VNH2SP30	5.5–16 V	30 A	Heavy duty, robust protection
Cytron MD10C	5–30 V	10 A	Clean PWM, great for this size

Recommended: L298N or BTS7960

12 GA DC Motor · Heavy-Duty Brushed

Driver IC	Voltage	Current	Notes
BTS7960 (IBT-2)	5.5–27 V	43 A peak	Best choice for high-current 12GA
VNH2SP30	5.5–16 V	30 A	Monster driver, strong protection
DRV8701	5.9–45 V	15 A	Industrial grade, reliable
Cytron MD10C	5–30 V	10 A	Clean PWM, easy to interface
Sabertooth 2x32	6–30 V	32 A / ch	Advanced features, regenerative braking

Recommended: BTS7960 or VNH2SP30

Micro BLDC Motor · Brushless DC

Driver IC	Voltage	Current	Notes
ESC (Generic)	depends	depends	Plug & play, simplest integration
DRV10983	1.5–10 V	2 A	Sensorless BLDC driver IC, dedicated
STSPIN230	1.8–10 V	1.3 A	Micro BLDC, perfect for tiny motors
TMC6300	2–11 V	300 mA	Ultra low voltage BLDC driver
A4964	7–50 V	—	Sensorless BLDC control IC
DRV8313	8–60 V	2.5 A / ph	3-phase BLDC driver, compact

Recommended: ESC (easiest) or STSPIN230 / DRV10983

3 · Quick Reference Summary

Motor	Best Driver	Difficulty	Use Case	Verdict
N20 Gear Motor	TB6612FNG / DRV8833	Easy	General / beginner	★ Best Overall
Coreless Motor	TB6612FNG / DRV8833	Medium	Speed competition	★ Best for Speed
16 GA Motor	L298N / BTS7960	Easy	Mid-weight bots	Good for larger builds
12 GA Motor	BTS7960 / VNH2SP30	Medium	Heavy / industrial	Overkill for line follower
Micro BLDC	ESC / STSPIN230	Hard	Advanced competition	Expert builds only

